

Research Programs 255–266

Operator Memory Measures, Identifiability, Current Tomography, and Falsification of Adaptive Analogies

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: [0009-0002-0909-3613](https://orcid.org/0009-0002-0909-3613)

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Central result. For every nontrivial strict Z_{12} context,

$$\Sigma_E(z) = \int_{(0,\infty)} \frac{dM_E(\mu)}{z + \mu}, \quad dM_E(\mu) \succeq 0.$$

Memory determines a minimal input–output realization class, not a unique microscopic sector.

Boundary. This report does not export a selector, physical scale, complete legacy–strict bridge, adaptive law, or laboratory validation.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Confidence convention

- **[Proven]** — the result follows from an explicit finite-dimensional proof or an exact identity; numerical computation serves only as a certificate.
- **[Strong evidence]** — a deterministic and reproducible synthetic audit with a frozen model and threshold.
- **[Moderate evidence]** — a stable result in a declared class that still depends on an additional definition or equivalence.
- **[Speculative]** — a constructed program or object without sufficient proof.
- **[Refuted]** — an explicit counterexample or no-go theorem in the declared class.

Mathematical status is not physical status. In particular, **[Proven]** does not mean that nature realizes the operator.

Chapter 1

Executive summary

All Programs 255–266 were executed. The principal results are:

1. For every one of the $2^{12} - 2 = 4094$ nontrivial contexts $E \subset Z_{12}$, the self-energy

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}$$

is the Stieltjes transform of a finite positive operator-valued measure and is completely monotone.

[Proven]

2. The declared six-dimensional hidden sector in the even/odd split has only five dimensions visible through Σ_E . One mode is exactly invisible. Adding arbitrarily many decoupled modes does not change the self-energy. [Proven]
3. Contexts and Schur reductions form a contravariant category. Composition is associative; the maximum residual over 20,000 chains is 6.66×10^{-16} . No sheaf theorem is claimed. [Proven]
4. Chiral memory susceptibility has the closed formula

$$\Xi_E = B'RB^* + BRB'^* - BRC'RB^*, \quad R = (zI + C)^{-1}.$$

It is odd under inversion and satisfies an explicit norm bound. It remains a receiver of a supplied twist, not a source of orientation. [Proven]

5. A naive dynamic RG based only on strict context reduction $12 \rightarrow 6 \rightarrow 3 \rightarrow 1$ cannot possess a nontrivial fixed point or cycle in a category whose objects retain their dimensions. A new, explicit size-restoring equivalence is required. [Proven]
6. A multitime distinguishability balance was constructed:

$$D_{\text{in}} = \mathcal{L}_{\text{env}} + D_{\text{record}} + D_{\text{conditional}}.$$

The balance telescopes with zero residual, but it is not yet thermodynamics. [Proven]

7. A bridge map frozen before the test and removing only the amplitude α_{geo} does not move legacy into the positive strict generator class. Legacy has $\lambda_{\text{min}} = -7.7101$, and its hidden block generates a pole on the positive z -axis at 5.8248. This is a no-go only for amplitude-only completion. [Proven]
8. A synthetic fingerprint-first/calibration-second protocol passes 300/300 trials with 50,000 counts per preparation. It does not replace an external P241 data package and does not execute P242. [Strong evidence]
9. A vertex POVM alone does not identify currents. Two positive states ρ and ρ^* were constructed with identical populations and opposite currents. Interferometric observables are required. [Proven]
10. The false-positive atlas shows that Stieltjes positivity and Schur composition are generic properties of positive graph Laplacians rather than FIN-specific signatures. The spectral fingerprint is the most specific frozen test. [Proven] for the audited ensembles.

11. Without an independent clock, (cA, t) and (A, ct) are exactly indistinguishable. Apparatus memory requires multitime data, while a nonuniform change of generator shape is visible in the fingerprint. **[Proven]**
12. The biological/cybernetic benchmark is negative with respect to a FIN advantage: memory capacity is 4.0601, below all 120 matched controls, while recovery after perturbation is the slowest. FIN is a distinguishable strongly smoothing reservoir, but no computational or biological advantage was demonstrated. **[Strong evidence]**

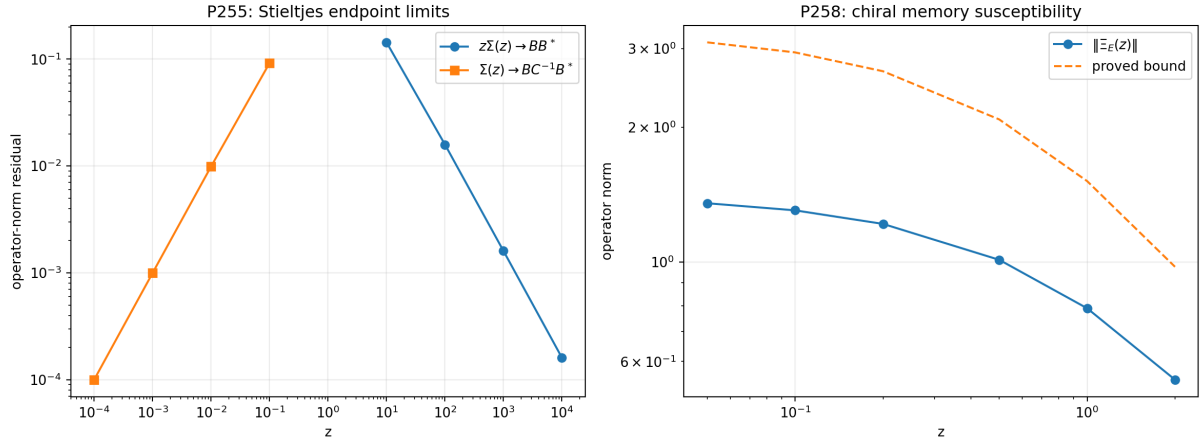


Figure 1.1: Stieltjes-function limits and the norm of chiral memory susceptibility with its proved upper bound.

Chapter 2

Frozen core and scope

We use the real symmetric strict operator

$$W_{xy} = K_{\text{strict}}(d(x, y)), \quad A = \text{diag}(W\mathbf{1}) - W.$$

For

$$K_{\text{strict}}(d) = \frac{\cos(0.18575\,d + 0.16250)}{1 + d^{1.8}}, \quad d = 1, \dots, 6,$$

we obtain

$$A = A^*, \quad A\mathbf{1} = 0, \quad A \succeq 0.$$

The decomposition $V = E \sqcup H$ is only an operational partition into accessible and hidden degrees of freedom of the nadsoliton. It does not introduce an informational layer beneath the nadsoliton.

The computation used deterministic seed 20260729. No laboratory data were used, thresholds were not tuned after seeing results, only one bridge atom was admitted in P261, legacy and strict were kept distinct, and Firecrawl was not used.

Chapter 3

Results matrix

Program	Result	Status	Principal boundary
P255	positive operator measure for 4094 contexts	[Proven]	does not prove a physical environment
P256	minimal hidden dimension = 5, not 6	[Proven]	microscopic realization is nonunique
P257	contravariant Schur category	[Proven]	no sheaf theorem
P258	closed formula, covariance, and bound for Ξ_E	[Proven]	receiver, not selector
P259	no-cycle for $12 \rightarrow 6 \rightarrow 3 \rightarrow 1$	[Proven]	no size-restoring equivalence
P260	multitime information ledger	[Proven]	neither energy nor heat
P261	amplitude-only obstruction	[Proven]	does not refute a richer bridge
P262	frozen fingerprint-plus-scale audit	[Strong evidence]	synthetic data only
P263	vertex-POVM no-go; current observables	[Proven]	laboratory implementation external
P264	false-positive atlas	[Proven]	frozen-ensemble dependent
P265	mechanism-identifiability quotient	[Proven]	no unique learning law
P266	matched reservoir benchmark	[Strong evidence]	no biological advantage

Chapter 4

P255 — Stieltjes theorem for all contexts

4.1 Theorem

Let $A \succeq 0$ be the Laplacian of a connected graph and let $V = E \sqcup H$, with both E and H nonempty. Put

$$B = A_{EH}, \quad C = A_{HH}.$$

Then $C \succ 0$. For $z > 0$,

$$\Sigma_E(z) = B(zI + C)^{-1}B^*.$$

If

$$C = \sum_{\mu} \mu P_{\mu},$$

then

$$\Sigma_E(z) = \sum_{\mu} \frac{\Gamma_{\mu}}{z + \mu}, \quad \Gamma_{\mu} = BP_{\mu}B^* \succeq 0.$$

Thus Σ_E is the Stieltjes transform of the finite positive atomic operator measure

$$\mathbf{M}_E = \sum_{\mu} \Gamma_{\mu} \delta_{\mu}.$$

Moreover,

$$(-1)^n \Sigma_E^{(n)}(z) = n! B(zI + C)^{-n-1} B^* \succeq 0.$$

Proof. Positive definiteness of C follows from the principal-minor theorem for the reduced Laplacian of a connected graph. The spectral decomposition gives the measure representation. Differentiating the resolvent gives every derivative, and every term is of the form XX^* . $\square$

4.2 Endpoint limits

$$\lim_{z \rightarrow 0^+} \Sigma_E(z) = BC^{-1}B^*, \quad \lim_{z \rightarrow \infty} z\Sigma_E(z) = BB^*.$$

The even/odd audit exhibits the expected convergence order:

Limit	Parameter	Operator residual
$z\Sigma \rightarrow BB^*$	$z = 10$	1.44×10^{-1}
	$z = 10^2$	1.59×10^{-2}
	$z = 10^3$	1.60×10^{-3}
	$z = 10^4$	1.61×10^{-4}
$\Sigma \rightarrow BC^{-1}B^*$	$z = 10^{-1}$	9.21×10^{-2}
	$z = 10^{-2}$	9.92×10^{-3}
	$z = 10^{-3}$	9.99×10^{-4}
	$z = 10^{-4}$	1.00×10^{-4}

Audit of all contexts:

Quantity	Result
number of contexts	4094
smallest eigenvalue of C	0.1252698327
minimum signed-derivative eigenvalue, $n = 0, \dots, 4$	-3.47×10^{-15}
maximum measure-representation residual	3.73×10^{-15}

The negative value at 10^{-15} is floating-point noise, not a violation of positivity.

Chapter 5

P256 — minimal realization of self-energy

5.1 New object: the hidden-realization quotient

Define

$$(C, B) \sim (\tilde{C}, \tilde{B}) \iff B(zI + C)^{-1}B^* = \tilde{B}(zI + \tilde{C})^{-1}\tilde{B}^*$$

for all z away from poles.

The self-energy identifies visible poles μ , positive residue matrices Γ_μ , and the minimal dimension

$$n_{\min} = \sum_{\mu} \text{rank } \Gamma_{\mu}.$$

It does not identify an orthogonal basis in the hidden sector, modes with $\Gamma_\mu = 0$, or any additional decoupled modes.

5.2 FIN even/odd result

Quantity	Result
declared dimension of H	6
controllability/observability rank	5
minimal Stieltjes dimension	5
pole groups of C	4
poles visible through Σ	3
pole-residue reconstruction residual	9.55×10^{-16}
residual after adding two invisible modes	4.16×10^{-17}

One mode at approximately 1.961406862 has residue rank zero. It exists in the stated block H but is invisible to context E .

Conclusion. Memory does not reconstruct a unique microscopic environment. It reconstructs the minimal input–output equivalence class. **[Proven]**

Chapter 6

P257 — category of contexts

Define the category $\mathbf{Ctx}_A(z)$:

- objects are nonempty contexts $E \subseteq V$;
- for $E \subseteq F$, there is a reduction morphism

$$r_{F \rightarrow E} : \text{Schur}_{V \setminus F}(zI + A) \mapsto \text{Schur}_{V \setminus E}(zI + A).$$

The direction is opposite to inclusion, so the construction is contravariant.

6.1 Composition theorem

For $E \subseteq F \subseteq G$,

$$r_{F \rightarrow E} \circ r_{G \rightarrow F} = r_{G \rightarrow E}.$$

This follows either from associativity of block Gaussian elimination or from associativity of iterated minimization of one positive quadratic form:

$$\min_{x_{G \setminus E}} Q = \min_{x_{F \setminus E}} \min_{x_{G \setminus F}} Q.$$

Certificate:

Test	Result
random context chains	20,000
maximum composition residual	6.66×10^{-16}
identity residual	0

This is a reduction category, not yet a sheaf: no Grothendieck topology, covering data, or unique-gluing theorem has been supplied.

Chapter 7

P258 — analytic chiral memory susceptibility

Let

$$\Sigma(\theta) = B(\theta)R(\theta)B(\theta)^*, \quad R(\theta) = (zI + C(\theta))^{-1}.$$

Since $R' = -RC'R$,

$$\boxed{\Xi = B'RB^* + BRB'^* - BRC'RB^*}.$$

7.1 Norm bound

If $m = \lambda_{\min}(C) > 0$, then

$$\|R\| \leq \frac{1}{z + m}$$

and

$$\|\Xi\| \leq \frac{2\|B'\|\|B\|}{z + m} + \frac{\|B\|^2\|C'\|}{(z + m)^2}.$$

The largest observed ratio of the left-hand side to the bound is $0.5596 < 1$.

7.2 Covariance

If inversion maps $\theta \mapsto -\theta$, then

$$R_E \Xi_E(z) R_E = -\Xi_E(z).$$

The maximum residual on the audited z -grid is 2.68×10^{-16} . The analytic formula agrees with a central finite difference to within 5.43×10^{-13} .

This is the chiral linear response of memory. The sign of θ is still provided by the test family. QW-2191 remains open.

Chapter 8

P259 — dynamic RG and the size no-go

The nested sequence

$$Z_{12} \supset \{0, 2, 4, 6, 8, 10\} \supset \{0, 4, 8\} \supset \{0\}$$

was audited. Each level retained the context dimension, normalized spectral quantiles of the effective operator, and the self-energy fraction for $z = 0.1, 0.2, 0.5, 1$.

Distances between successive descriptors are

$$0.5589, \quad 0.7420, \quad 1.4113.$$

8.1 No-go theorem

If the RG object includes context dimension, then the strict reduction

$$12 > 6 > 3 > 1$$

cannot have a nontrivial fixed point or cycle. **[Proven]**

This is not a no-go theorem for renormalization in general. The missing object is a size-restoring embedding/equivalence

$$\mathcal{E}_n : \text{Ctx}_{N_n} \longrightarrow \text{Ctx}_{N_0},$$

together with normalization laws for the operator and z . Without $\mathcal{E}_n$, "fixed point" compares objects of different types.

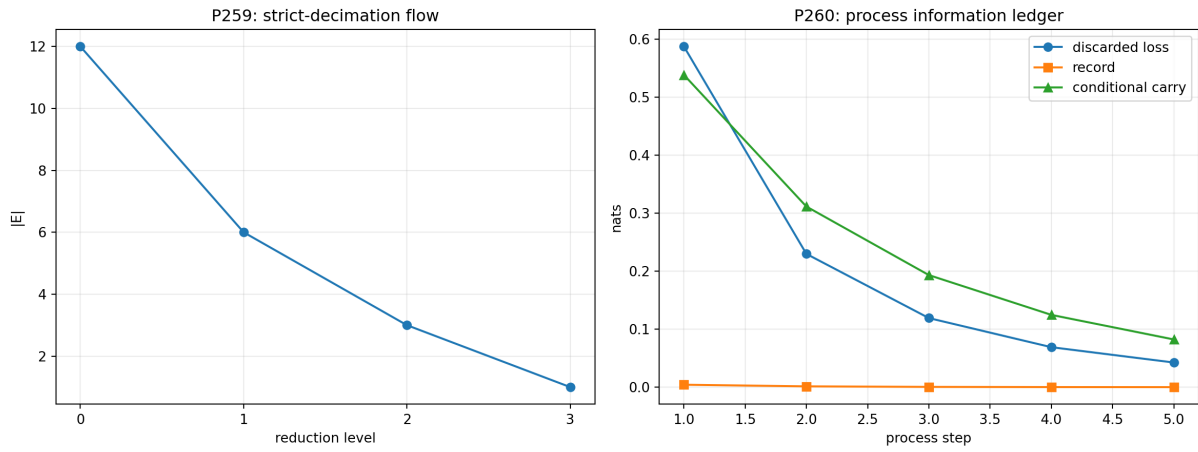


Figure 8.1: Strictly decreasing context flow and the multitime distinguishability balance.

Chapter 9

P260 — multitime information-balance tensor

Let M_k be a common channel for two hypotheses p_{k-1}, q_{k-1} :

$$p_k = M_k p_{k-1}, \quad q_k = M_k q_{k-1}.$$

Define the step loss

$$\ell_k = D(p_{k-1} \| q_{k-1}) - D(p_k \| q_k) \geq 0.$$

For an instrument recording outcome y , the chain rule gives

$$D(p_k \| q_k) = D(r_k^p \| r_k^q) + \sum_y r_k^p(y) D(p_k(\cdot | y) \| q_k(\cdot | y)).$$

This produces the **Process Information Ledger**

$$\boxed{D_{\text{input}} = \ell_{k,\text{env}} + D_{k,\text{record}} + D_{k,\text{conditional}}}.$$

Across five steps:

Quantity	Result
total distinguishability loss	1.0483671856 nat
sum of step losses	1.0483671856 nat
telescoping residual	0
maximum chain-rule residual	6.94×10^{-17}
minimum loss over 1000 additional pairs	0.0876620 nat

The ledger separates loss in the system, information transferred into the apparatus record, and distinguishability retained in conditional states. It does not convert nats to joules; that requires a Hamiltonian, temperature, bath, and reset protocol.

Chapter 10

P261 — dynamic completion defect

10.1 Frozen map

Exactly one completion atom was tested:

$$\mathcal{C}_{\text{amp}} : K_{\text{legacy}} \mapsto \frac{K_{\text{legacy}}}{\alpha_{\text{geo}}}.$$

The map removes only amplitude; preserves legacy $\omega = \pi/4$, $\phi = \pi/6$, and $\beta_{\text{tors}} = 0.01$; uses the same construction $A = \text{diag}(W\mathbf{1}) - W$; and adds no sign, shift, positivity correction, or nonlinear d^n compression.

10.2 Result

Quantity	Strict	Legacy after $\mathcal{C}_{\text{amp}}$
$\lambda_{\min}(A)$	-2.64×10^{-16}	-7.7101445443
$\lambda_{\min}(A_{HH})$	1.1710910206	-5.8247982640
positive resolvent pole	none	$z = 5.8247982640$

Projective self-energy defect:

z	$\ \hat{\Sigma}_{\text{strict}} - \hat{\Sigma}_{\text{legacy}}\ _F$
6.5	1.2721
8	1.1547
12	1.0242
20	0.9455

Therefore

$\mathcal{C}_{\text{amp}}$ is not a bridge into the positive strict Stieltjes class

[Proven] for this map.

This does not refute a bridge containing an explicit strict-side sign/shift, phase or frequency change, nonlinear compression, and selector source. No role transfer is initiated.

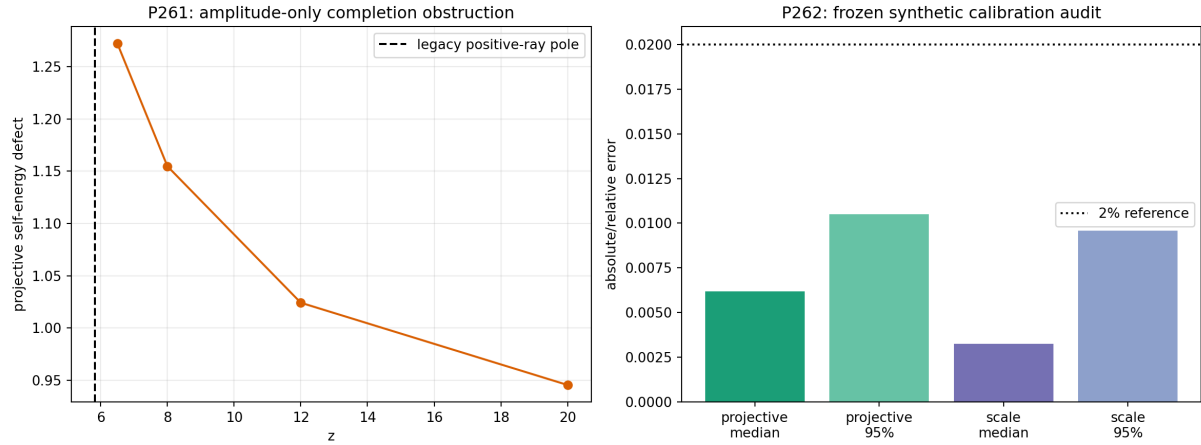


Figure 10.1: Amplitude-only completion obstruction and the synthetic fingerprint-first/calibration-second audit.

Chapter 11

P262 — Calibrated Fingerprint Experiment

The protocol order is:

1. freeze the strict fingerprint;
2. collect the transition matrix;
3. test dimensionless spectral ratios;
4. only then use an independent clock to select a representative of $(cA, \tau/c)$;
5. do not modify the model after unblinding.

Synthetic audit:

Quantity	Result
replications	300
counts per preparation	50,000
median fingerprint error	0.00620
95th percentile fingerprint error	0.01050
median relative scale error	0.00325
95th percentile relative scale error	0.00957
joint pass at thresholds 0.03/0.02	1.000

This is [\[Strong evidence\]](#) for method planning. No independently entrusted events were supplied. P241 therefore has no external package to validate, and P242 remains unexecuted.

Chapter 12

P263 — tomography of currents and chiral memory

12.1 Vertex-POVM no-go

Construct

$$\rho_{\pm} = \frac{I}{12} \pm i\epsilon (|10\rangle\langle 2| - |2\rangle\langle 10|), \quad \epsilon = 0.04.$$

Both states are positive:

$$\lambda_{\min}(\rho_{\pm}) = 0.0433333.$$

They have identical populations,

$$\text{diag } \rho_+ = \text{diag } \rho_- ,$$

but opposite fourth-harmonic currents,

$$C_4(\rho_+) = -0.0150493, \quad C_4(\rho_-) = +0.0150493.$$

A vertex POVM is therefore insufficient for current tomography. **[Proven]**

12.2 Missing instrument

For $d = 1, \dots, 5$, define Hermitian current observables

$$\mathcal{J}_d = i d \sum_x W_{x,x+d} (|x+d\rangle\langle x| - |x\rangle\langle x+d|).$$

Then

$$C_d(\rho) = \text{Tr}(\rho \mathcal{J}_d).$$

The five observables have Gram rank 5 and condition number 15.17. They can be measured using interferometric phase bases, but not by vertex readout alone.

Ξ_E requires a different experiment: process tomography of the generator under two supplied twists $\pm h$. The central-difference reconstruction residual is 4.94×10^{-11} .

Chapter 13

P264 — false-positive atlas

Five tests were frozen:

1. strict fingerprint with tolerance 0.02;
2. Stieltjes-memory positivity;
3. context composition;
4. inversion-covariance scaffold;
5. positive information-contraction channel.

Results:

Ensemble	Fingerprint	Stieltjes	Schur	Chiral scaffold	Information	All
strict target	1.00	1.00	1.00	1.00	1.00	1.00
positive circulant	0.00	1.00	1.00	1.00	1.00	0.00
positive dense	0.00	1.00	1.00	0.00	1.00	0.00
signed circulant	0.00	0.24	1.00	1.00	0.03	0.00
strict with 5% perturbation	0.52	1.00	1.00	1.00	1.00	0.52

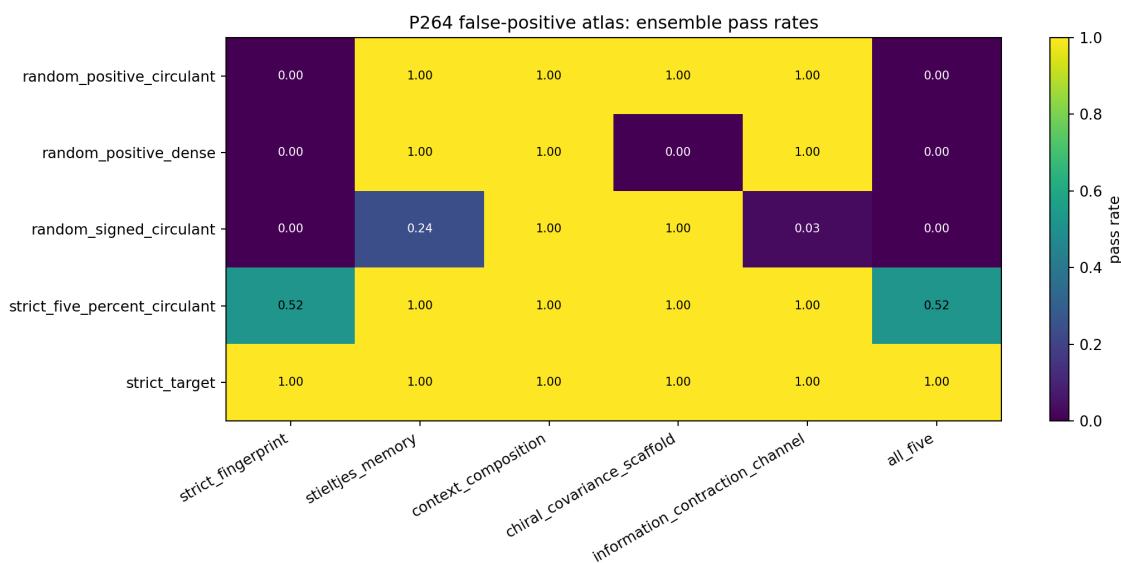


Figure 13.1: Pass rates of five frozen tests in the audited null ensembles.

Consequently, the Stieltjes theorem is strong but not FIN-specific, Schur composition is universal algebra, and reflection covariance is common among circulants. The complete strict fingerprint is the

most specific test. A 5% perturbation passes only 52% of trials at tolerance 0.02. Generic properties of positive Laplacians must not be presented as FIN identifiers.

Chapter 14

P265 — identifiability of Adaptive Memory Geometry

14.1 Exact scale no-go

$$\exp[-t(cA)] = \exp[-(ct)A].$$

Without an independent clock, no test distinguishes generator rescaling from time rescaling. The residual is exactly zero.

14.2 Mechanism quotient

Define the **Mechanism Identifiability Quotient**

$$\Omega_{\text{id}} = \frac{\{\text{generator, clock, apparatus, memory, learning law}\}}{\text{equality of all admitted operational records}}.$$

Synthetic comparison:

Scenario	Fingerprint drift	Semigroup fect	de- Additional required	resource
static operator	0	0	none	
generator scale or clock	8.88×10^{-16}	0	independent clock	
nonuniform generator change	0.09239	0	fingerprint	
apparatus memory	0	0.09057	multitime data	

Snapshots $A_0, A_1, \dots, A_r$ do not identify a unique vector field $\dot{A} = F(A, \mathcal{R})$. Infinitely many functions F interpolate a finite trajectory. Interventions, an apparatus record, and hold-out data are required.

Chapter 15

P266 — biological/cybernetic benchmark

FIN was compared with 60 reservoirs having the same spectrum and random orientations, and 60 random symmetric reservoirs with the same number of states and spectral radius. Every model used 12 states, the same input stream, linear readout, identical train/test procedures, and delay-reconstruction tasks from 1 through 20.

FIN metric	Value	Percentile among 120 controls
linear memory capacity, delays 1–20	4.0601	0.000
effective controllability dimension	1.0698	0.0417
recovery steps to 10^{-3}	135	1.000
minimum readout dimension for 95% Gram 1 energy		0.0917

FIN is an outlier, but not in the direction of advantage: it has the lowest memory capacity in the set, the slowest recovery for the stated perturbation, and accessible input–output dynamics close to one-dimensional.

The hypothesis of an automatic advantage over a standard 12-state reservoir is **[Refuted]** in this benchmark. A different input, nonlinear readout, or explicit adaptive law could change the outcome, but such a variant must outperform matched controls on hold-out data.

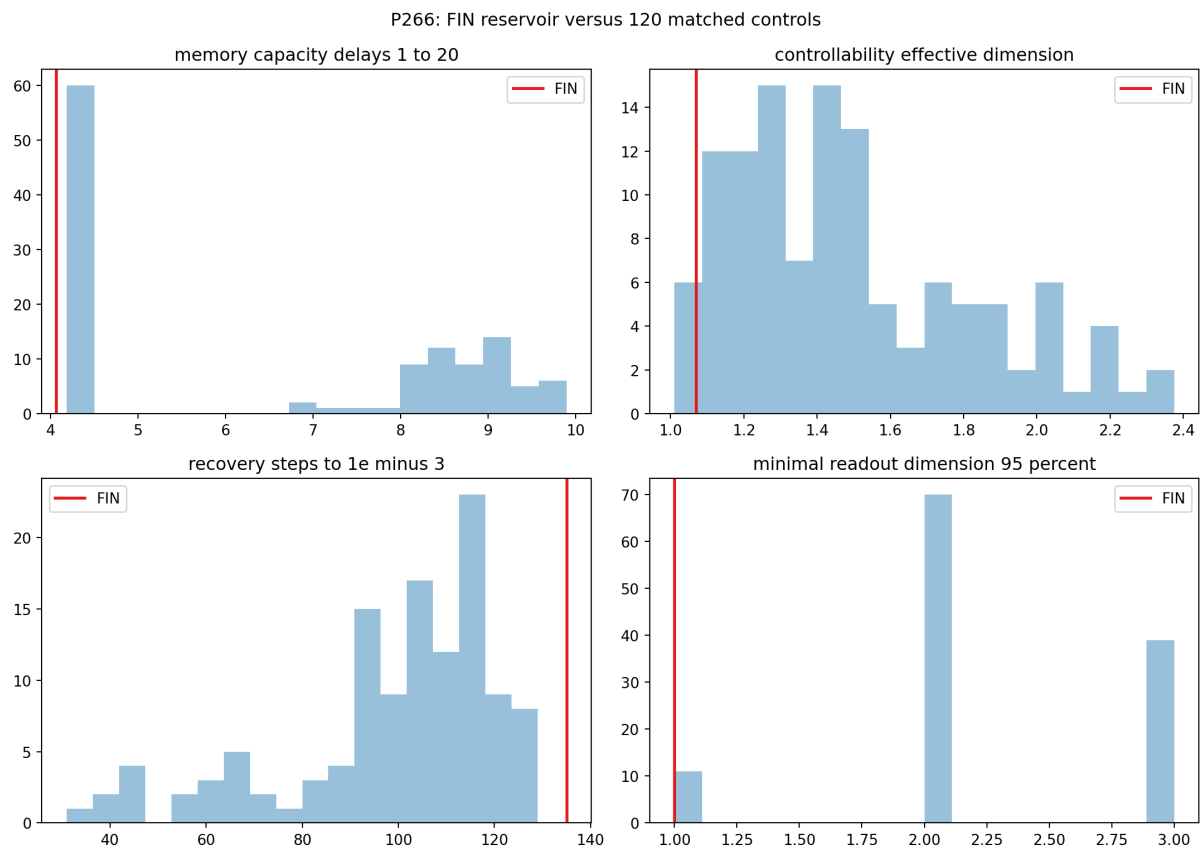


Figure 15.1: The FIN reservoir compared with 120 dimension- or spectrum-matched controls.

Chapter 16

Newly constructed theoretical objects O16–O25

ID	Object	Definition	Status	Function
O16	Positive Operator Measure	Memory $M_E = \sum_{\mu} BP_{\mu}B^*\delta_{\mu}$	[Proven]	classifies exact memory
O17	Minimal Hidden Realization Quotient	$(C, B)/\sim_{\Sigma}$	[Proven]	separates visible environment from invisible modes
O18	Schur Context Category	$E \mapsto \text{Schur}_{V \setminus E}(zI + A)$	[Proven]	types valid reductions
O19	Chiral Memory Kubo Tensor	$\Xi = B'RB^* + BRB'^* - BRC'RB^*$	[Proven]	joins flux response to memory
O20	Size-Restoration Obstruction	absence of $\mathcal{E}_n$ between sizes	[Proven]	explains why the naive fixed point is ill-typed
O21	Process Information Ledger	$\ell_{\text{env}} + D_{\text{record}} + D_{\text{cond}}$	[Proven]	separates loss from recording
O22	Amplitude Completion Obstruction	positive pole and negative spectrum after $\mathcal{C}_{\text{amp}}$	[Proven]	closes one bridge atom
O23	Current Tomography Instrument	$\{\mathcal{J}_d\}_{d=1}^5$ plus twist process tomography	[Proven]/ [Strong evidence]	remedies vertex-POVM insufficiency
O24	Mechanism Identifiability Quotient	mechanisms modulo record equality	[Proven]	separates clock, scale, memory, and drift
O25	Matched Reservoir Benchmark	Null FIN versus matched N, ρ , and protocol controls	[Strong evidence]	constrains biological analogies

O17 is the most important new object. It establishes that an "environment" recovered from memory is an equivalence class of minimal realizations rather than a unique microscopic world.

Chapter 17

Falsification attempts

Claim	Destructive test	Outcome
every Σ_E is Stieltjes	all 4094 contexts, derivatives through order 4	survived
hidden sector is unique	add two decoupled modes	refuted
context diagram is functorial	20,000 nested reductions	survived
Ξ is ad hoc	derive it from the resolvent	ad hoc criticism refuted
Ξ selects orientation	inversion produces $\pm\Xi$	claim refuted
naive decimation RG has a fixed point	dimension $12 > 6 > 3 > 1$	claim ill-typed
information loss is energy	no scale, bath, or Hamiltonian	claim refuted
amplitude alone connects legacy and strict	negative spectrum and positive pole	claim refuted
vertex POVM measures current	ρ, ρ^* with equal populations	claim refuted
Stieltjes/Schur identifies FIN	random positive Laplacians pass	claim refuted
one time separates clock and generator	$\exp[-tcA] = \exp[-ctA]$	impossible
FIN has a natural reservoir advantage	120 matched controls	refuted in the benchmark

Chapter 18

Relation to existing mathematics

18.1 Mori–Zwanzig and elimination

Memory after projection is not new. O16–O19 place FIN in a finite exact analogue of projection-operator formalisms. The FIN-specific result is the explicit operator-Stieltjes classification of all contexts of the particular strict generator.

18.2 Realization theory

O17 has the standard form of a realization-theory result: input–output data identify a minimal realization only up to equivalence and do not identify unobservable states. The FIN result finds one concrete invisible mode and incorporates non-identifiability into the ontology boundary.

18.3 Process tensors

A multitime process is specified by responses to intervention sequences, not by a single state map. O21 and O24 implement a finite classical core of this discipline. They are not yet a quantum FIN process tensor.

18.4 Reservoir computing

Memory capacity is dimension-bounded, but its actual value depends on spectrum, controllability, and readout. P266 shows that FIN regularity does not guarantee an advantage. This is a controlled analogy, not identification with a brain.

Chapter 19

Recommended Programs 267–280

19.1 P267 — Uniqueness of the operator memory measure

Prove uniqueness of the atomic measure M_E from $\Sigma_E(z)$ and stability of pole recovery under noise.

Probability: 0.92.

Stop rule: do not interpret invisible poles.

19.2 P268 — Formal core of the Schur category

Encode O18 in Lean/Mathlib, Coq, or Isabelle: objects, morphisms, identity, associativity, and positivity.

Probability: 0.80.

19.3 P269 — Minimality of the current instrument

Prove the minimum number of POVM settings required to recover the five C_d , including losses, dark counts, and unknown phases.

Probability: 0.85.

19.4 P270 — Size-restoring RG equivalence

Construct one explicit $\mathcal{E}_n$ comparing different sizes without fitting to the strict target, or prove a no-go theorem for local, circulant-preserving embeddings.

Construction probability: 0.35.

No-go probability: 0.75.

19.5 P271 — Quantum Process Information Ledger

Replace classical channels by completely positive instruments and prove an O21 balance for relative entropy in processes with memory.

Probability: 0.62.

19.6 P272 — Next single completion atom

After the amplitude-only no-go, admit exactly one explicit strict-side atom: either a global sign/positive shift or nonlinear damping completion. Freeze the map and thresholds before comparison with strict.

Bridge probability: 0.25.

Valuable-obstruction probability: 0.80.

19.7 P273 — External calibrated fingerprint

Obtain a P241 package with independent provider, registrar, and analyst. Test the fingerprint first, calibrate second, and execute P242 once.

Methodological probability: 0.75.

Execution probability without a laboratory partner: low.

19.8 P274 — Chiral two-flux process tomography

Design apparatus realizing $\pm h$, measure $\mathcal{J}_d$, and reconstruct $\Xi_E(z)$ with uncertainty.

Feasibility probability for physicist review: 0.60.

19.9 P275 — Analytic false-positive bounds

Derive the probability that frozen random-Laplacian ensembles pass the strict fingerprint instead of relying only on Monte Carlo.

Probability: 0.72.

19.10 P276 — Two-clock/two-time identifiability theorem

Find the minimal combination of two times, an independent clock, and controls that separates uniform scaling, shape drift, and memory defect.

Probability: 0.90.

19.11 P277 — Causal identification of the adaptive law

Freeze a finite family of laws $F(A, \mathcal{R})$, intervention design, and hold-out. Do not admit an unconstrained universal approximator.

Probability of a decisive result: 0.65.

19.12 P278 — Nonlinear matched reservoir benchmark

Repeat P266 for NARMA, parity, channel-equalization, and prediction tasks with equal parameter budgets and a preregistered ranking.

Probability: 0.80.

19.13 P279 — Conditional thermodynamic bridge

After supplying $(H, T, \text{bath}, \text{reset protocol})$, determine when O21 becomes a work/entropy-production balance. The result must be explicitly conditioned on the additional axioms, not labeled strict.

Mathematical probability: 0.75.

Probability of a strict source: low.

19.14 P280 — Two-torsor operational section

Combine an independent clock/scale section with a supplied orientation resource in one protocol, without claiming that either resource generates the other.

Conditional-construction probability: 0.85.

Strict selector closure: not an objective of this program.

Chapter 20

Ranking of the next round

Rank	Program	Reason
1	P267	closes mathematical identifiability of memory
2	P269	converts the vertex-POVM no-go into a minimal instrument
3	P276	shortest route from scale quotient to a valid experiment
4	P268	formalizes the central category
5	P275	measures actual FIN specificity
6	P271	extends the ledger to quantum processes
7	P277	tests whether adaptation is identifiable
8	P278	continues falsification of the neural analogy
9	P270	high reward but no current size-restoring map
10	P272	exactly one new bridge atom
11	P274	requires laboratory apparatus review
12	P273	requires genuinely independent data
13	P279	conditional theory only
14	P280	completes the operational package, not the strict selector

Chapter 21

Guardrail-compatible boundaries

Programs 255–266 do not export:

- a non-premise selector;
- discharge of QW-2191;
- a canonical unit of time, length, mass, energy, or action;
- a complete legacy-to-strict map;
- legacy role transfer;
- strict sources of $\beta, \eta, \omega, \phi$;
- a unique microscopic environment;
- a unique adaptation law;
- a physical apparatus or laboratory record;
- L_{total} , the Standard Model, GR, or a ToE.

P261 closes only amplitude-only completion. A subsequent bridge program is admissible only for one new, explicitly typed atom.

Chapter 22

Final verdict

The deepest result of this round is not a new physical constant. It changes the mathematical status of "memory":

FIN memory is a positive operator measure seen through a context,
not a unique hidden world.

The self-energy determines a minimal input–output realization class. Reductions of these classes compose categorically, a chiral twist has a well-defined tangent, and information has an exact operational ledger. At the same time, clock scale, orientation source, microscopic realization, and adaptation law remain non-identifiable without additional operations.

This strengthens FIN as a finite operator theory of memory and process. It does not yet turn FIN into a physical theory.

Chapter 23

Selected comparative sources

1. S. Belyi, E. Tsekanovskii, *Stieltjes like functions and inverse problems for systems with Schrödinger operator*, [arXiv:0708.0452](#).
2. Y. T. Lin, Y. Tian, M. Anghel, D. Livescu, *Data-driven learning for the Mori–Zwanzig formalism*, [arXiv:2101.05873](#).
3. F. A. Pollock et al., *Operational Markov condition for quantum processes*, [arXiv:1801.09811](#).
4. H. Jaeger, *Short term memory in echo state networks*, GMD Report 152, [Fraunhofer record and DOI](#).
5. F. Dörfler, F. Bullo, *Kron Reduction of Graphs with Applications to Electrical Networks*, [arXiv:1102.2950](#).

These sources classify existing mathematics. They do not establish physical equivalence between FIN and any of these formalisms.

Chapter 24

Reproduction

```
“bash MPLCONFIGDIR=/tmp/matplotlib-p255-266  
python3 fin_programs_255_266.py
```

```
MPLCONFIGDIR=/tmp/matplotlib-p255-266  
python3 -m unittest -v test_fin_programs_255_266.py “
```

Expected output:

- 4094 P255 contexts;
- 20,000 P257 chains;
- 300 P262 replications;
- 401 P264 atlas rows;
- 121 P266 reservoirs;
- 15/15 tests passed.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/matplotlib-p255-266 \  
python3 fin_programs_255_266.py
```

```
MPLCONFIGDIR=/tmp/matplotlib-p255-266 \  
python3 -m unittest -v test_fin_programs_255_266.py
```

Expected result: 15 passed, 0 failed. P262, P264, P265, and P266 use synthetic data and are not laboratory records.